

Reform and Practice of Teaching Mode for “Principles and Applications of Microcontrollers” in the Perspective of Digital and Intelligent Transformation

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Abstract

In the context of the rapid development of digitalization and intelligentization, the traditional teaching mode of the “Principles and Applications of Microcontrollers” course is facing challenges such as slow update of teaching content, unclear progression of abilities, difficulty in achieving personalized learning, weak process evaluation, and difficulty in evaluating the achievement of abilities. This paper proposes a new course teaching mode for the course, which is a four-fusion, three-progressive, and two-combination model. Based on the OBE concept as the overall design, it deeply integrates digitalization empowerment methods such as knowledge graphs, ability graphs, AI tutors, and digital virtual humans. In terms of teaching content, it further promotes the integration of production and education, science and education, competition and innovation, and innovation and entrepreneurship in the course. In terms of teaching methods, it innovates the deep blended teaching method based on ability progression of BOPPPS + ENGINE. In terms of teaching evaluation, it constructs a new evaluation system that combines process evaluation and Terminal evaluation, and dynamic tracking and precise feedback. The practical results show that the students in the experimental group have significantly improved their ability to solve complex engineering problems, verifying the feasibility and effectiveness of the teaching model proposed in this paper in the digitalization context for course reform, and providing a practical path for improving the teaching quality of similar courses that can be widely promoted.

CCS Concepts

• **General and reference** → Document types; Reference works.

Keywords

Digital and intelligent transformation, Teaching mode, Microcontrollers, OBE

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1 Introduction

With the rapid development of technologies such as the Internet of Things, artificial intelligence, and big data, the global industrial landscape is undergoing profound changes, and the demand for applied and innovative talents in engineering education is becoming increasingly urgent. The “China Education Modernization 2035” clearly states the reform direction of “strengthening the cultivation of practical abilities”, and the “New Engineering Education Construction Action Plan” of the Ministry of Education also emphasizes the urgency of “reconstructing the curriculum system based on industrial demands”. The “Key Points of the Ministry of Education’s Work in 2022” clearly proposes “implementing the national education digitalization strategic action”, aiming to promote the comprehensive digital transformation of the education sector. With the penetration of technologies such as generative AI in the education field, traditional programming courses have gradually exposed weak effects in formative evaluation, making it difficult to comprehensively and dynamically reflect students’ knowledge mastery and ability improvement during the learning process. Against this backdrop, the teaching pain points of traditional programming courses, such as slow update of teaching content, unclear progression of abilities, difficulty in achieving personalized learning, weak process evaluation, and difficulty in evaluating the achievement of abilities, have become increasingly prominent, and are unable to meet the students’ needs for the ability to solve complex engineering problems[1].

Therefore, this paper proposes a new four-fusion, three-progressive, and two-combination course teaching mode. With the OBE concept as the overall design, it deeply integrates knowledge graphs, ability graphs, AI tutors, digital virtual humans and other digital and intelligent empowerment methods. In terms of teaching content, it further promotes the integration of production and education, science and education, competition and innovation, and innovation and entrepreneurship in the course. In terms of teaching methods, it innovates the deep blended teaching method based on the progressive development of abilities, namely BOPPPS + ENGINE. In terms of teaching evaluation, it constructs a new course evaluation system that combines process evaluation and Terminal evaluation, as well as dynamic tracking and precise feedback.

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Through empirical research in the “Principles and Applications of Microcontrollers” course, this teaching model is verified to be feasible and effective in enhancing students’ ability to solve complex engineering problems, providing theoretical support and practical references for the improvement of teaching quality of similar courses in the context of digital and intelligent transformation.

2 The main teaching problems to be addressed

The digital and intelligent transformation has brought about a series of significant changes in education and teaching. The teaching of the “Principles and Applications of Microcontrollers” course is confronted with numerous issues and challenges such as the update of teaching content, the optimization of teaching methods, and the improvement of teaching evaluation. These are specifically manifested in the following aspects.

2.1 The industrial demands change rapidly while the teaching content updates slowly

With the widespread application of generative AI tools such as DeepSeek and ChatGPT, the demand for professional innovative talents’ job capabilities in enterprises has changed rapidly, and it has shifted from traditional manual content generation to the debugging and verification of AI-generated content. The traditional teaching model lacks an effective mechanism to update teaching content in a timely and dynamic manner based on industrial demands, and thus cannot meet the ability requirements of enterprises for professional innovative talents in the digital era.

2.2 The progression of abilities is unclear and individualized learning is difficult to achieve

Although digital, intelligent and information technologies have provided abundant resources and means for course teaching, due to the unclear delineation of the levels of students’ ability advancement, the lack of in-depth understanding and precise grasp of individual differences among students, it is impossible to provide customized learning paths and personalized guidance support for each student. As a result, students’ learning needs and interests cannot be fully met, and there are still many obstacles to achieving personalized learning.

2.3 The process evaluation is weak and the feedback on ability achievement is hard to provide

With the widespread use of generative AI tools, students are extensively using AI to generate answers for their process-based evaluation tasks. When teachers deal with a large number of students, it is difficult for them to conduct detailed observation and evaluation of each student’s evaluation process. Teachers also struggle to distinguish the difference between “copying code” and “self-debugging”, which leads to students having a utilitarian tendency of “emphasizing results over process”. This further exacerbates the weakness of process-based evaluation. As a result, traditional process-based evaluation cannot accurately reflect students’ ability achievement based on the evaluation results.

3 The establishment of a new course teaching mode

3.1 Overall framework design of the teaching model

A four-fusion, three-progressive and two-combination course teaching model under the background of digitalization and intelligence has been proposed (see Figure 1). This model takes the OBE concept as the overall design, deeply integrates digital and intelligent empowerment methods such as knowledge graphs, ability graphs, AI tutors, and digital virtual humans, and conducts comprehensive and in-depth reforms and innovations in teaching content, teaching methods, and teaching evaluation.

3.2 Application of digital and intelligent tools

Numerous digital and intelligent tools such as knowledge graphs, ability graphs, smart teaching platforms, AI tutors, and digital virtual person have been widely adopted. Knowledge graphs have constructed a comprehensive and systematic knowledge system for microcontrollers, clearly presenting the connections and hierarchies among knowledge points, helping students build a complete cognitive framework[2]. Ability graphs clearly define the various abilities that students should achieve during the learning process, providing clear guidance for students’ personalized learning to enhance their abilities. By leveraging the smart education platform, analyzing students’ learning data, and precisely pushing personalized learning paths and resources. AI tutors can provide real-time online answers, promptly resolving students’ doubts, and can also automatically grade homework, significantly improving teaching efficiency. Courses utilize generative artificial intelligence to write MOOC scripts for the main knowledge points of each module, and then use digital virtual humans to quickly generate MOOC video resources. These resources can better and more promptly meet the personalized learning needs of students, providing rich and high-quality learning support, and promoting the continuous improvement of course quality and effectiveness.

Taking the knowledge graphs and ability graphs as examples, this article explains the specific applications of digital and intelligent tools and how they provide strong support for personalized learning for students. During the teaching process, the online intelligent education platform “Rain Classroom” of Tsinghua University was fully utilized. It can automatically construct the knowledge graphs based on the course contents. Considering the learning paths and ability development patterns of students, teachers further establish the ability graphs based on the knowledge graphs. Students can clearly see each ability level they need to reach during the learning process through the ability graphs. The platform can intelligently recommend suitable learning resources and exercises based on the current learning progress and ability level of students, thereby achieving personalized learning support. During the continuous update of course contents, the knowledge graphs and ability graphs will also be dynamically updated and maintained, ensuring the accuracy and timeliness of the knowledge graphs and ability graphs.

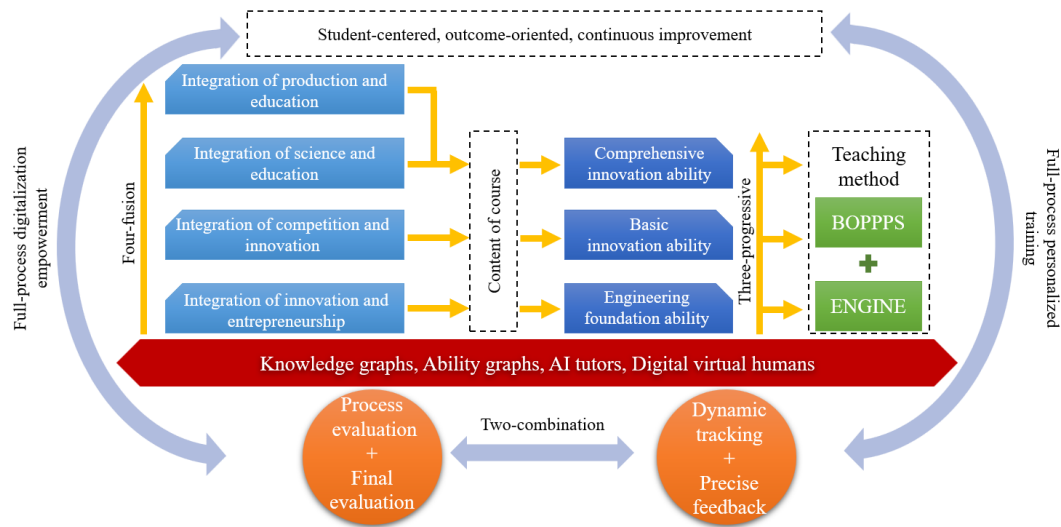


Figure 1: The new teaching mode proposed in this paper.

3.3 Reconstruction of course objectives based on OBE

Based on the investigation of the job requirements of small and medium-sized enterprises in Shaanxi Province and in accordance with the “Engineering Education Accreditation Standards (2020 Edition)”, the specific course objectives of this course are as follows:

(1) Master the level characteristics, I/O ports, internal memory, etc. of the MCS-51 series single-chip microcontroller and its hardware structure and principles, and be able to design the hardware of the single-chip microcontroller application system for the measurement and control requirements of mechanical systems.

(2) Master the program control methods of timers/counters, interrupt systems and serial communication, and be able to propose reasonable experimental plans for the measurement and control requirements of mechanical systems, and analyze and evaluate the feasibility of the measurement and control schemes centered on the single-chip microcontroller through experiments.

(3) Master the principles and methods of using C51 language and Keil software, master the program control methods of common interface circuits, and be able to program and debug the common interface circuits for the measurement and control schemes of mechanical systems.

3.4 Teaching content update based on the four-fusion

Through the four-fusion mechanisms of production and education integration, science and education integration, competition and innovation integration, and innovation and entrepreneurship integration, the teaching content can be updated rapidly to keep up with the demands of industrial development in the digital and intelligent era. Through production and education integration, we deeply align with the actual needs of enterprises, integrate cutting-edge industry technologies and typical application cases into the teaching content, transform them into teaching resources, update the

course content, promote the implementation of case-based teaching, and enhance the practicality and employment orientation of the courses. Through science and education integration, we introduce teachers’ research achievements and the latest technological trends, broaden students’ horizons, and cultivate their research thinking and innovation capabilities. Through competition and innovation integration, we encourage students to integrate the concept of innovation and entrepreneurship into their professional studies, set up innovative practice modules in the teaching content, guide students to carry out single-chip microcomputer-related project development, and enhance their innovation awareness and ability. Through innovation and entrepreneurship integration, we use discipline competitions as a carrier, convert the requirements of competition projects, technical difficulties, etc. into teaching content, stimulate students’ enthusiasm for exploration, and hone their ability to solve complex engineering problems. Through the four integration mechanisms, the teaching content can be updated rapidly, laying a solid foundation for students to adapt to the development of industries in the digital and intelligent era.

3.5 Innovative approach of BOPPPS + ENGINE blended teaching method based on the three-progressive

An innovative BOPPPS + ENGINE blended teaching method based on three-progressive has been developed. This method proposes a three-level ability achievement model, focusing on the step-by-step cultivation of engineering foundation skills, basic innovation skills, and comprehensive innovation skills. Engineering foundation skills serve as the cornerstone, providing students with a solid professional foundation and enabling them to master the basic principles and operational skills in fields such as microcontrollers; basic innovation skills guide students to break away from conventional thinking and attempt to apply the knowledge they have learned to solve general engineering problems; comprehensive innovation

skills require students to complete innovative project design and implementation in complex engineering scenarios.

In each stage of the teaching process for cultivating various abilities, this method integrates the BOPPPS model and the ENGINE model deeply. The BOPPPS model consists of six components: Bridge-in, objective, pre-assessment, participatory learning, post-assessment, and summary[3]. With clear instructional design, it ensures the precise achievement of teaching goals and the optimization of teaching effects from an engaging introduction to a conclusive summary. The ENGINE model comprises six components: case introduction, knowledge analysis, problem exploration, skill training, innovation integration, and engineering practice. It is a closed-loop teaching system oriented towards engineering practice, driven by problems, and aiming at enhancing abilities, achieving the integration of theoretical teaching and practical application. Combined with the BOPPPS model, the ENGINE model plays a significant role in the participatory learning and post-assessment stages, promoting the occurrence of deep learning. This blended teaching method is guided by the three-progressive ability model, achieving the organic combination of blended teaching and project-based teaching, effectively enhancing students' innovative and practical abilities in solving complex engineering problems, and effectively promoting the cultivation of applied innovative talents.

3.6 Establishment of a diversified evaluation system based on the two-combination

A diversified evaluation system has been established, empowered by digital and intelligent technologies, featuring multi-dimensional, dynamic tracking, and precise feedback. Through AI tools, comprehensive collection, mining, and analysis of students' learning data are carried out, forming multi-dimensional evaluation perspectives, enabling real-time monitoring and dynamic tracking of students' learning progress, learning behaviors, and learning outcomes[4, 5], and providing timely and accurate personalized feedback on the achievement of phased abilities for students, promoting the continuous improvement of their individual abilities. The assessment methods and evaluation system design for this course are as follows:

(1) Process evaluation (60%)

This includes MOOC learning (5%), graphs learning (5%), homework (10%), online tests (10%), and experiments (30%). Among them, homework and online tests are conducted according to knowledge modules, using the intelligent education platform and knowledge graphs to generate personalized test questions to achieve precise assessment. Six engineering practice projects are set up, and through experimental operation assessment and the quality of experimental reports, the students' ability to solve complex engineering problems is comprehensively evaluated.

(2) Terminal evaluation (40%)

Terminal evaluation includes the final exam (20%) and project works (20%). The final exam is conducted in a closed-book format, comprehensively evaluating the students' achievement of knowledge and abilities. The project works are completed in a project team collaboration form for the designated single-chip microcomputer application system design, comprehensively evaluating the students' ability to solve complex engineering problems.

4 Practical Proof

4.1 Experimental subjects and cycles

Four natural classes (a total of 166 students) from the mechanical design and manufacturing and automation major of our school in the 2022 grade were selected as the experimental subjects. Among them, two natural classes (a total of 82 students) were the experimental group and adopted the new teaching mode; the other two natural classes (a total of 84 students) were the control group and adopted the traditional teaching mode. The experimental period was the first semester of the 2024-2025 academic year (16 weeks), covering all contents of the "Principles and Applications of Micro-controllers" course (32 hours of theory + 16 hours of experiments).

In order to scientifically evaluate the effectiveness of the new teaching model, we conducted a detailed statistical analysis of the learning outcomes of the experimental group and the control group. In terms of sample size, the sample sizes of the experimental group and the control group were 82 and 84 respectively. The sample sizes of the two groups were similar and thus comparable. In terms of the control group conditions, the control group adopted the traditional teaching mode, and was consistent with the experimental group in terms of teaching resources, teaching time, teaching environment and assessment schemes, in order to ensure the fairness of the comparison.

4.2 The calculation and results of the achievement values for the course objectives

4.2.1 Establish an evaluation relationship. The student's achievement value is calculated through a diversified evaluation system with weighted calculations. The process evaluation score is composed of the scores from MOOC learning, graphs learning, homework, online tests, experiments, etc. The terminal evaluation score is composed of the scores from the final exam and project works. For each evaluation method, clear, measurable, and easy-to-operate scoring criteria have been established, and the supporting relationship between various evaluation methods and course objectives has been clarified (see Table 1).

4.2.2 The calculation of achievement values. The calculation of the achievement values mainly employs the performance analysis method and the scoring sheet method. For MOOC learning, graphs learning, homework, online tests, and final exams, the performance analysis method is adopted. For experiments and project works, the scoring sheet method is used[6].

(1) Performance Analysis Method

The course objective i is supported by N evaluation points. Let the total score of the j th evaluation point be S_j , the average score of students be A_j , and the weight be ω_j . Then, the achievement value OA_i of course objective i is calculated according to the following formula:

$$OA_i = \sum_{j=1}^N \frac{\omega_j A_j}{S_j} \quad (1)$$

(2) Scoring sheet method

The scoring sheet method sets detailed, specific and measurable evaluation indicators as the basis for evaluation, sets different satisfaction levels and corresponding scores, forms a scoring table, and

Table 1: The supporting relationships between the evaluation methods and the course objectives, as well as the distribution of scores.

<i>Course objectives</i>	<i>Process evaluation</i>					<i>Terminal evaluation</i>		<i>Total</i>
	<i>MOOC learning</i>	<i>Graphs learning</i>	<i>Homework</i>	<i>On-line tests</i>	<i>Experiments</i>	<i>Final exam</i>	<i>Project Works</i>	
Objective 1	1	1	2	2	6	4	4	20
Objective 2	1.5	1.5	3	3	9	6	6	30
Objective 3	2.5	2.5	5	5	15	10	10	50
Total	5	5	10	10	30	20	20	100
	60					40		

Table 2: The results of achievement values.

<i>Groups</i>	<i>Course objectives</i>	<i>Achievement values</i>							<i>Objective achievement values</i>
		<i>MOOC Learning</i>	<i>Graphs Learning</i>	<i>Home-work</i>	<i>On-line tests</i>	<i>Experiments</i>	<i>Final exam</i>	<i>Project Works</i>	
Experimental group	Objective 1	0.726	0.737	0.753	0.712	0.774	0.751	0.803	0.763
	Objective 2	0.731	0.725	0.763	0.723	0.768	0.759	0.801	0.764
	Objective 3	0.767	0.716	0.775	0.734	0.792	0.758	0.791	0.772
	Objective 1	0.738	0.726	0.764	0.718	0.712	0.730	0.731	0.727
	Objective 2	0.730	0.714	0.773	0.726	0.718	0.725	0.725	0.728
	Objective 3	0.764	0.721	0.760	0.704	0.729	0.651	0.727	0.715

evaluates the achievement of course objectives. The scoring table adopts a five-level system, namely: excellent (A), good (B), medium (C), pass (D), and fail (E). The corresponding scores are 95 points, 85 points, 75 points, 65 points, and 55 points. If a student does not participate in the evaluation, the score is recorded as 0. Based on the completion of each evaluation method by the students, grades are evaluated, and after summarization and statistics, weighted calculations are made to obtain the achievement values of each course objective.

4.2.3 The results of achievement values. Using the performance analysis method and the scoring sheet method, the achievement values of the course objectives for the experimental group and the control group were calculated. The results are shown in Table 2. The expected achievement value for this course was set at 0.7. Therefore, all the course objectives of both the experimental group and the control group were achieved.

4.3 Analysis of Practical Effect

In terms of the achievement values of the course objectives, there was a significant difference between the experimental group and the control group. From the specific values, the achievement values of the experimental group in all three course objectives were higher than those of the control group. Taking Objective 1 as an example, the achievement value of the experimental group was 0.763, while that of the control group was 0.727, indicating that the experimental group exceeded the control group by approximately 4.95%. In Objective 2, the achievement value of the experimental group was 0.764, and that of the control group was 0.728, with the experimental group exceeding by approximately 4.94%. On Objective 3, this

gap was even more significant, with the achievement value of the experimental group being 0.772 and that of the control group being 0.715, indicating that the experimental group exceeded the control group by approximately 7.97%.

Through an independent samples t-test, we compared the differences in the achievement of course objectives between the experimental group and the control group. The results showed that, on all three course objectives, the average score of the experimental group was significantly higher than that of the control group ($p < 0.05$). Specifically, on objective 1, the average score of the experimental group was 0.763, while that of the control group was 0.727, with a t-value of 2.34 and a p-value of 0.021; on objective 2, the average score of the experimental group was 0.764, while that of the control group was 0.728, with a t-value of 2.29 and a p-value of 0.024; on objective 3, the average score of the experimental group was 0.772, while that of the control group was 0.715, with a t-value of 3.12 and a p-value of 0.002.

To assess the actual effectiveness of the new teaching model, we also calculated Cohen’s d effect size. The effect size for course objective 1 was 0.62, for course objective 2 it was 0.58, and for course objective 3 it was 0.75. On all three course objectives, the effect size was greater than 0.5, indicating that the new teaching model has a moderately positive impact on students’ learning outcomes.

5 Conclusion

This paper proposes a new course teaching mode for “Principles and Applications of Microcontrollers” course that integrates four-fusion, three-progressive, and two-combination. This mode effectively utilizes artificial intelligence technology to empower education and teaching, providing assistance in teaching, learning, management,

and evaluation. It aims to explore and promote the application of digital and intelligent tools, promoting precise teaching, personalized learning, and diversified evaluation. The practical application of the “Principles and Applications of Microcontrollers” course shows that the students in the experimental group achieved significantly better results in meeting the course objectives compared to the control group, especially in experiments and project works. This indicates that the new teaching mode proposed in this paper can significantly enhance students’ ability to solve complex engineering problems. It is feasible and effective in the digital and intelligent context for the reform of the “Principles and Applications of Microcontrollers” course, and can also provide a replicable practical path for improving the teaching quality of similar courses.

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The intelligent course construction projects of Xijing University: ZHKC2501. The Project of Teacher Education Reform and

Teacher Development Research of Shaanxi Province in China: SJS2025YB049.

References

- [1] Ye Qing, Li Ze, Ma Haiying. 2025. Exploration of Teaching Reform in “Principles and Applications of Microcontrollers” under the Background of Engineering Education Professional Certification. *Die & Mould Manufacture*, 2025, 25 (02): 89-91.
- [2] Wu Yelan, Yu Wanying, Zhang Wanying. 2024. Knowledge Graph Construction of Microcontroller Courses for Intelligent Education. *Computer and Information Technology*, 2024, 32(06):87-91.
- [3] Huang Yan. 2025. Research on the Application of OBE+BOPPPS Blended Teaching Mode in the Perspective of Digital Transformation—Taking Database Course Groups as an Example. *Information & Computer*, 2025, 37 (04): 176-178.
- [4] D. Ifenthaler, J.Y.K. Yau. 2020. Utilising learning analytics to support study success in higher education: a systematic review. *Educ. Technol. Res. Dev.*, 2020, 68 (4):1961–1990.
- [5] Bakhouyi Abdellah, Dehbi Amine, Amhaimar Lahcen. 2025. Improving smart learning: course completion via AI-driven hybrid system integration in big data. *Telematics & Informatics Reports*, 2025, (18):1-10.
- [6] Li Ze, Zhang Yanzhong, Chu Peng. 2020. The Evaluation of the Objectives Achievement Degree for “Principles and Application of Single-chip Microcomputer” Course Based on OBE Model. 2020 International Conference on Information Science and Education (ICISE-IE2020), Sanya, China, 2020:109-112.